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Eating disorders across population groups with different weight status in Belgium: evidence for an integrative preventative approach

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Abstract

Background It is a common misperception that eating disorders (ED) predominantly affect individuals with underweight. This study aims to clarify the prevalence and odds of living with an ED across different weight groups within the Belgian population, and to identify sociodemographic factors associated with the comorbidity of ED and overweight or obesity.

Methods Data for this study was sourced from the 2018 Belgian Health Interview Survey (BHIS). This is a cross-sectional survey that collects health information from approximately 10,000 individuals living in Belgium through face-to-face interviews and self-administered questionnaires. Data collection involved questions on EDs (SCOFF questionnaire), height and weight, and sociodemographic characteristics. The prevalence of EDs was determined through descriptive statistics. Multivariable logistic regression models were employed to assess the odds of living with an ED across different weight categories and to identify sociodemographic variables (age, sex, education, income, and ethnicity) linked to the comorbidity of EDs and obesity/overweight.

Results General ED prevalence in this study was 7.1%. The prevalence of EDs was highest among individuals with obesity (12.3%, 95% CI: 10.0–15.1%) compared to those with normal weight (5.5%, 95% CI: 4.4–6.9%). Individuals with obesity had three times higher odds of living with an ED (adjusted odds ratio (AOR): 3.0, 95% CI: 2.1–4.1). Individuals with overweight also had increased odds of living with an ED (AOR: 1.4, CI: 1.0–2.0). The comorbidity of EDs and obesity/overweight was significantly linked to low educational levels (AOR: 1.8, 95% CI 1.1–2.8), financial difficulties (AOR: 1.9, 95% CI 1.4–2.7), and maternal non-EU origin (AOR: 2.4, 95% CI 1.6–3.7).

Conclusions EDs disproportionately affect people with higher weight. This underscores the necessity for public health interventions that address the intertwined nature of EDs and obesity. Traditional weight-centric health campaigns focusing on individual responsibility may inadvertently contribute to the problem by promoting weight stigma. An integrative preventative approach holds promise for more effective management and prevention of EDs and obesity in the Belgian population.

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Keywords Eating disorders, Obesity, Prevalence, Sociodemographic factors, Comorbidity, Public health policy, Belgium

Introduction

Obesity and eating disorder (ED) prevention are two major public health priorities of our time, yet insufficient research has been conducted to address the complex way they interact. After a decades-long war waged against obesity [1, 2], reports show that the prevalence of overweight and obesity is only increasing [3, 4]. Not only do current treatments and public health campaigns seem to lack effectiveness, there is a growing concern about their potential collateral damage. Namely, multiple studies have reported that focusing on weight loss rather than health promotion might increase the risk for disordered eating [5–7].

EDs, including Anorexia Nervosa (AN), Bulimia Nervosa (BN), and Binge Eating Disorder (BED), are serious conditions affecting both mental and physical health. They are characterised by a disturbed relationship to food, eating, body image, and physical exercise [8–10]. EDs can lead to severe health problems and have the highest mortality rate among mental disorders [11, 12]. Disordered Eating Behavior (DEB) is a broader term where symptoms of an ED are present, but not frequently or severely enough to warrant psychiatric diagnosis [13].

Obesity and overweight are contested terms. Previously, they were solely defined based on Body Mass Index (BMI), with cut-offs for overweight and obesity set at a BMI of ≥ 25 and ≥ 30 respectively [1]. This perspective has evolved, with obesity now defined as an abnormal or excessive fat accumulation that affects health [14]. Having overweight, but especially obesity, is associated with multiple chronic conditions like cardiovascular disease, diabetes, musculoskeletal disorders, and some cancers [3].

Historically, EDs were misperceived to primarily affect people who have underweight [15, 16]. Substantial evidence from the past few decades has since revealed that individuals with a current or past higher body weight often suffer from EDs [17–21]. Early work by Stunkard (1959) was among the first to report on DEB in higher weight individuals, documenting cases of binge eating among those with obesity [22]. This behaviour was later formalized as BED by Spitzer et al. (1992) [22, 23]. Initially, higher body weight was mainly associated with EDs characterized by binge eating episodes, such as BN and BED [22–26]. In contrast, AN, defined in part by a significantly low body weight [8], is theoretically incompatible with higher weight. Nonetheless, research by Lebow et al. (2015) in the US demonstrated that patients with restrictive EDs like AN often have a history of higher body weight, including previous overweight or obesity [27].

For these patients, who meet the behavioural criteria for AN but not the weight criterion, the diagnosis of Atypical Anorexia Nervosa (AAN) is applied [8].

Both obesity and EDs share common risk factors, such as dieting, weight stigma (i.e. the social devaluation and denigration of individuals because of their excess body weight [28]), and body dissatisfaction [26, 29–34]. When EDs and obesity/overweight co-occur, their health consequences are compounded. Firstly, the numerous physical and mental adverse effects related to EDs and obesity interact synergistically. This results in a greater deleterious impact on health [35]. Secondly, individuals with a lifetime history of obesity experience ED symptoms that are more severe, last longer, and are more resistant to treatment [36, 37]. For these reasons, it is important to distinguish high risk groups.

Studies that have investigated sociodemographic factors related to the comorbidity of ED and obesity/overweight are scarce. One study by Darby et al. (2009) in Australia found that young females had the highest rate of comorbid higher weight and EDs [20]. Besides age and sex, no clear associations between sociodemographic status and the comorbidity of higher weight and EDs have been established. However, some evidence points to ethnicity and socioeconomic status (SES) as potential predictive or protective factors for comorbidity. For instance, in the US, Rodgers et al. (2017) found differences in the prevalence of DEB by ethnicity/race in a group of adolescents with overweight, but no consistent patterns of interaction emerged [38]. Concerning SES, low SES is consistently associated with higher prevalence of having obesity/overweight [39–41], whereas the relationship between SES and ED prevalence is less clear. EDs appear over a wide range of socioeconomic backgrounds [42]. A study by West et al. (2019) in the US found that having obesity/overweight was more strongly related to binge eating among adolescents from a high SES background, compared to those from a low SES background. Nevertheless, having obesity/overweight remained a risk factor for engaging in binge eating behaviour in both the high and low SES group [43]. Further exploration regarding potential sociodemographic factors is clearly needed, especially in European settings.

The present study aims to understand the association between weight status and EDs in Belgium. By analysing data from a representative nationwide health interview survey, the primary objective is to determine the odds of living with an ED across population groups with different weight status in the Belgian population. A second objective is to discern whether there are any specific

sociodemographic factors associated with the comorbidity of ED and obesity/overweight.

This paper argues that EDs manifest across the entire weight spectrum, but hypothesises that disordered eating behaviour is more common at a higher BMI. In addition, hypothetical risk factors for comorbid ED and obesity/overweight that are explored in this study are age, sex, educational level, income, and ethnicity.

Method

Data

Statistical analysis was performed using the 2018 dataset of the Belgian Health Interview Survey (BHIS): a cross-sectional survey performed every 4 to 5 years collecting health information from around 10,000 individuals through a face-to-face interview and a self-administered questionnaire [44]. In 2018 there were 11,611 participants. The individuals partaking in the BHIS were sampled from different Belgian municipalities using the National Register as a sampling frame. The survey methodology consisted of stratification and multistage cluster sampling. This sampling method, in combination with post-stratification weights, ensure that the findings are generalisable to the Belgian population. Further details about the survey methodology can be found in the paper by Demarest et al. (2013) [44].

The statistical analysis included information on age, sex, education, income, BMI, and ED risk.

In the survey, living with an ED was determined using the SCOFF questionnaire [45], an instrument validated to screen for probable EDs [46], specifically AN and BN [47]. It consists of five yes/no questions regarding weight loss, disordered eating behaviour and cognitions, and body image. The presence of an ED is presumed when two or more questions are answered affirmatively.

Questions in the SCOFF questionnaire:

1. Have you recently lost 6 kilos or more in a 3-month period?
2. Do you worry that you have lost control over how much you eat?
3. Do you make yourself sick because you feel uncomfortably full?
4. Do people say you are too thin while you believe yourself to be too fat?
5. Would you say that food dominates your life?

As EDs are regarded as a sensitive subject, the questions were placed in the self-administered questionnaire, and asked only to those older than 15 years of age.

The BMI indicator (kg/m^2) was calculated based on self-reported height and weight (asked in the face to face questionnaire), and categorised as underweight ($<18.5 \text{ kg/m}^2$), normal weight ($18.5\text{--}24.9 \text{ kg/m}^2$),

overweight ($25.0\text{--}29.9 \text{ kg/m}^2$), and obesity ($\geq 30 \text{ kg/m}^2$) [1]. Although BMI is a rough estimate for body fat in a clinical setting, it significantly correlates with body fat at the population level [48], and can therefore serve as a reliable screening tool in epidemiological studies. In children and adolescents, the cut-off points to define BMI categories are age and sex-specific, and are based on the cut-off values of the International Obesity Taskforce (IOTF) [49]. For this reason, only participants older than 18 years of age were included in the study.

Highest educational level and 'ability to make ends meet' of the household were used as estimates for SES, and applied to its individual members. Educational level is based on the highest education attained in the household, and was categorised into three levels: 'low' (no diploma or primary education, or only lower secondary education), 'intermediate' (higher secondary education), or 'high' (all degrees higher than secondary education). Household income was assessed by asking respondents about their ability to make ends meet with their available income, and responses were classified as either 'easily making ends meet' or 'having difficulty in making ends meet'. While subjective, this question has been shown to correlate with objective income levels and is a robust indicator of financial strain and material deprivation [50]. Additionally, the mother's country of birth was used to estimate ethnicity, categorizing individuals as having a mother that is 'born in Belgium', 'born in the European Union (EU)', or 'born outside the EU'. It is important to note that the survey question did not specify whether the mother was biological, and thus this variable reflects sociocultural background rather than genetics.

The final sample size consisted of 7,643 individuals, aged 18 to 101. There were 7,857 individuals older than 18 years of age who filled in the auto-questionnaire. Individuals were excluded from the analysis when there was missing information on the SCOFF-questionnaire, their weight status, or their mother's country of birth. Figure 1 displays a flow chart reflecting how the final sample size was attained.

Analysis

Firstly, a descriptive analysis was performed to determine prevalence of EDs among people in different weight categories. Secondly, a multivariable logistic regression model was used to assess sociodemographic variables associated with the odds of living with an ED. In addition, a second logistic regression model was constructed to investigate which sociodemographic variables were associated with comorbid ED and obesity/overweight. In our regression model, we included covariates based on both theoretical frameworks and empirical evidence from prior studies. Age and gender were included as covariates, as young females were established as a high-risk

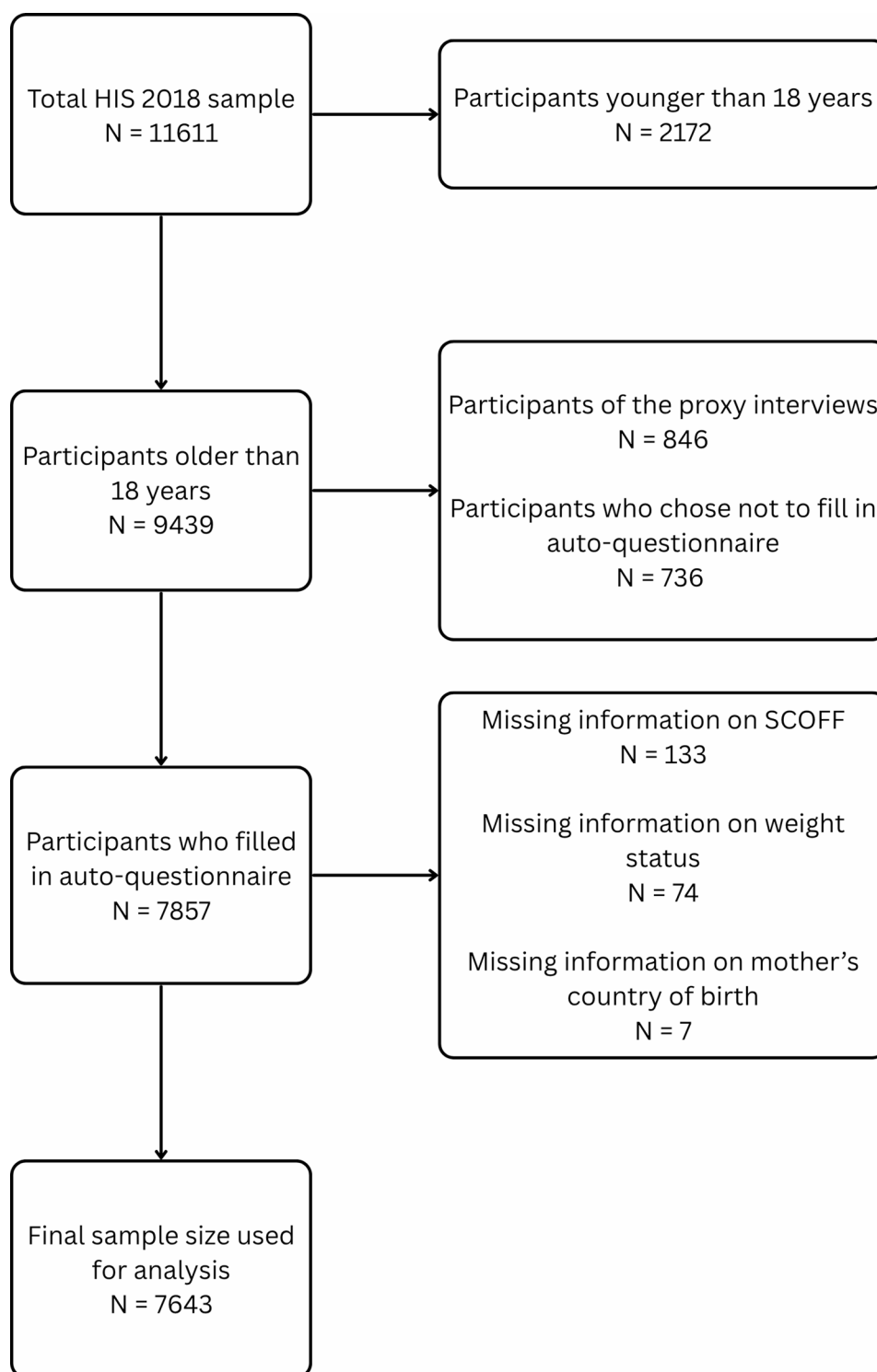


Fig. 1 Flow chart describing sample size

group [20]. Proxy variables for SES and ethnicity were also considered, since some studies suggest that SES and ethnicity play a role in the relationship between experiencing an ED and weight status [38, 43].

All analyses were weighted and have accounted for the complex study design of the BHIS. Confidence intervals were calculated at the 95% level. A p-value less than 0.05 was deemed significant. The analyses were performed in STATA 17 [51] using appropriate svy commands. The

STATA code will be shared upon request to the corresponding author.

Results

Population characteristics and ED prevalence by weight status

Out of the 7,643 participants, 7.1% were screened to live with an ED (Table 1). ED prevalence was higher at both ends of the weight spectrum, and those in the obesity category showed the highest prevalence out of all weight categories (12.3%, CI: 10.0–15.1%). Females had a higher prevalence of ED compared to males. A declining trend in ED prevalence by age group was observed, with the highest prevalence in the 18–24 year old group at 14.9% (CI: 10.9–20.1%), and the lowest in the 65+ year old group at 2.8% (CI: 2.1–3.8%). Educational level was inversely related to ED prevalence, as a low education showed the highest ED prevalence at 9.1% (CI: 7.1–11.4%). ED prevalence was also significantly higher in the group experiencing difficulty in making ends meet, at 12.5% (CI: 10.6–14.7%). Individuals with a mother born

outside the EU had the highest ED prevalence, at 17.1% (CI: 13.6–21.4%) (Table 1).

Odds of living with an ED by weight status

The multivariable logistic regression analysis (Table 2) revealed that the odds of living with an ED was three times higher (adjusted odds ratio (AOR): 3.0, CI: 2.1–4.1) for those in the obesity category compared to those with normal weight. For overweight and underweight the AORs were similar, at 1.4 (CI: 1.0–2.0) and 1.4 (CI: 0.8–2.8) respectively. However, the results were only significant for individuals with overweight compared to normal weight. No significant results were found for the underweight category.

Sociodemographic predictors of eds

Regarding sociodemographic characteristics, age and sex were significantly associated with living with an ED, as younger individuals and females showed increased odds. Additionally, a low educational level was associated with a 50% increase in the odds of living with an ED (AOR: 1.5, CI: 1.1–2.1), and those experiencing difficulty in

Table 1 Sociodemographic characteristics (N = 7643), weighted prevalence of eds and comorbidity of ED and obesity/overweight

	Weighted proportion	Weighted ED		Weighted comorbidity ED and obesity/ overweight	
	%	%	95% CI	%	95% CI
Sex					
Male	48.5	5.7	4.7–7.0	3.6	2.8–4.7
Female	51.5	8.4	7.3–9.6	4.8	4.0–5.8
Age (in years)					
18–24	8.0	14.9	10.9–20.1	5.1	3.0–8.7
25–44	32.4	8.9	7.4–10.6	5.3	4.2–6.7
45–64	36.4	6.5	5.3–7.9	4.7	3.7–6.0
65+	23.2	2.8	2.1–3.8	1.9	1.3–2.7
Educational level					
Low	17.6	9.1	7.1–11.4	5.7	4.2–7.6
Intermediate	32.5	8.0	6.5–9.8	5.0	3.8–6.5
High	48.6	5.6	4.7–6.6	3.0	2.4–3.8
No response or missing	1.2	13.3	6.0–34.0	12.6	4.0–33.1
Making ends meet					
Difficulty making ends meet	28.8	12.5	10.6–14.7	2.9	2.4–3.6
Easily making ends meet	69.7	4.8	4.1–5.7	7.5	6.0–9.4
No response or missing	1.5	8.3	3.9–16.9	3.7	1.2–10.8
Mother's country of birth					
Born in Belgium	79.5	5.6	4.9–6.5	3.3	2.7–4.0
Born in the EU	9.9	8.1	6.1–10.7	4.6	3.2–6.5
Born outside the EU	10.6	17.1	13.6–21.4	11.2	8.2–15.1
BMI category					
Underweight	2.8	9.2	5.1–16.0	-	-
Normal Weight	47.4	5.4	4.5–6.6	-	-
Overweight	33.6	6.7	5.4–8.2	-	-
Obesity	16.3	12.3	10.0–15.1	-	-
Total	-	7.1	6.3–7.9	4.3	3.7–5.0

Table 2 Logistic regression analysis for EDs

	Unadjusted			Adjusted		
	OR	95% CI	p-value	AOR	95% CI	p-value
Sex						
Male	1.0 (referent)	-	-	1.0 (referent)	-	-
Female	1.5	1.2–1.9	0.002	1.6	1.2–2.1	0.001
Age (in years)						
18–24	6.1	3.8–9.8	< 0.001	7.7	4.7–12.7	< 0.001
25–44	3.4	2.3–4.9	< 0.001	3.7	3.4–5.5	< 0.001
45–64	2.4	1.7–3.5	< 0.001	2.4	1.6–3.6	< 0.001
65+	1.0 (referent)	-	-	1.0 (referent)	-	-
Educational level						
Low	1.7	1.2–2.3	0.001	1.5	1.1–2.1	0.023
Intermediate	1.5	1.1–2.0	0.008	1.2	0.9–1.6	0.303
High	1.0 (referent)	-	-	1.0 (referent)	-	-
No response or missing	3.1	1.1–8.9	0.038	2.0	0.7–5.3	0.185
Making ends meet						
Difficulty making ends meet	2.8	2.2–3.6	< 0.001	2.0	1.5–2.7	< 0.001
Easily making ends meet	1.0 (referent)	-	-	1.0 (referent)	-	-
No response or missing	1.8	0.8–4.1	0.165	1.2	0.5–2.9	0.663
Mother's country of birth						
Born in Belgium	1.0 (referent)	-	-	1.0 (referent)	-	-
Born in the EU	1.5	1.1–2.1	0.023	1.3	0.9–1.9	0.117
Born outside the EU	3.5	2.5–4.8	< 0.001	2.2	1.6–3.1	< 0.001
BMI category						
Underweight	1.8	0.9–3.4	0.096	1.4	0.8–2.8	0.269
Normal Weight	1.0 (referent)	-	-	1.0 (referent)	-	-
Overweight	1.3	0.9–1.7	0.145	1.4	1.0–2.0	0.023
Obesity	2.5	1.8–3.4	< 0.001	3.0	2.1–4.1	< 0.001

making ends meet had twice the odds (AOR: 2.0, CI: 1.5–2.7) of living with an ED. Furthermore, individuals with a mother born outside the EU had more than twice the odds of living with an ED (AOR 2.2, CI: 1.6–3.1).

Sociodemographic predictors of comorbid ED and obesity/overweight

The second logistic regression model (Table 3) investigated sociodemographic variables associated with comorbid ED and obesity/overweight. The odds of living with an ED and simultaneously having overweight or obesity were significantly higher in those with a lower educational level (AOR: 1.8, CI: 1.1–2.8), those experiencing difficulty making ends meet (AOR: 1.9, CI: 1.4–2.7), and those who had a mother from a non-European country (AOR: 2.4, CI: 1.6–3.7). Female sex was not found to be associated with odds of comorbid ED and obesity/overweight (AOR: 1.4, 1.0–1.9). All age categories had lower odds of comorbidity compared to the 65+ age group, but odds did not differ significantly between the age categories below 65+.

Discussion and recommendations

The objective of this study was twofold: first, we aimed to evaluate the odds of living with an ED across different weight statuses, and second, we intended to explore sociodemographic characteristics associated with comorbid ED and obesity/overweight, utilising a representative nationwide sample from Belgium from 2018. As anticipated, this study established that the prevalence of EDs is highest among individuals with obesity, at 12.3%. Moreover, the study has found strong evidence that individuals with obesity also have the highest odds of living with an ED (AOR 3.0, CI: 2.1–4.1). This aligns with results from previous studies. Duncan et al. (2017) reported highest odds of a lifetime ED in the obesity category, specifically an OR of 2.79 for women and 3.77 for men [52]. Al Banna et al. (2021) and Nagata et al. (2018) also found highest odds of DEB and ED in the obesity category, at 3.57 and 2.45, respectively [13, 17].

The findings of our study thus reinforce existing evidence that EDs occur more often in those with obesity/overweight [17–20], and highlight the importance of screening for EDs in people with a higher weight [27, 53, 54].

This cross-sectional analysis cannot confirm causality or directionality due to the absence of temporality in the

Table 3 Logistic regression analysis for comorbid ED and obesity/overweight

	Unadjusted			Adjusted		
	OR	95% CI	p-value	AOR	95% CI	p-value
Sex						
Male	1.0 (referent)	-	-	1.0 (referent)	-	-
Female	1.3	1.0–1.9	0.076	1.4	1.0–1.9	0.069
Age (in years)						
18–24	2.8	1.4–5.6	0.003	2.6	1.3–5.3	0.007
25–44	2.9	1.9–4.7	< 0.001	2.8	1.7–4.5	< 0.001
45–64	2.6	1.6–4.1	< 0.001	2.6	1.6–4.2	< 0.001
65+	1.0 (referent)	-	-	1.0 (referent)	-	-
Educational level						
Low	1.9	1.3–2.8	0.001	1.8	1.1–2.8	0.010
Intermediate	1.7	1.2–2.4	0.005	1.5	1.0–2.2	0.034
High	1.0 (referent)	-	-	1.0 (referent)	-	-
No response or missing	4.6	1.3–16.1	0.017	3.0	0.9–9.7	0.065
Making ends meet						
Difficulty making ends meet	2.7	2.0–3.7	< 0.001	1.9	1.4–2.7	< 0.001
Easily making ends meet	1.0 (referent)	-	-	1.0 (referent)	-	-
No response or missing	1.3	0.4–4.1	0.696	1.0	0.3–3.2	0.994
Mother's country of birth						
Born in Belgium	1.0 (referent)	-	-	1.0 (referent)	-	-
Born in the EU	1.4	0.9–2.2	0.121	1.2	0.8–1.9	0.396
Born outside the EU	3.7	2.5–5.5	< 0.001	2.4	1.6–3.7	< 0.001

data. Prior research, however, does suggest that EDs like BN often follow a history of having obesity/overweight, and in turn, disordered eating behaviour can lead to weight gain in the long term [55, 56]. As mentioned in the introduction, EDs and obesity/overweight share common risk factors [29–34], a prominent one being weight stigma [57]. The likelihood of being subjected to weight stigma increases with BMI [31], and weight stigma predicts future weight gain [30, 58, 59], creating a vicious cycle. Longitudinal studies by Pearlman et al. (2023), and Fortes et al. (2024) found that weight stigma was also a predictor for DEB and EDs [60, 61]. The differential experience of weight stigma could possibly contribute to sociodemographic differences in ED prevalence. Women are more likely to be discriminated against based on weight than men [13, 28, 62, 63], and, as noted above, weight stigma increases with higher BMI [31]. These both parallel the ED prevalence found in our study.

Despite the frequent co-occurrence of ED and obesity/overweight, little research has been devoted to sociodemographic factors associated with this comorbidity [20]. Therefore it is difficult to compare our results with those of previous studies. For example, young females are the only subpopulation previously identified as being at higher risk for this comorbidity [20]. Surprisingly, the present study did not find a significant result for younger age, nor for female sex. Elderly (65+), however, were less likely to have comorbid ED and obesity/overweight.

SES had a clear impact on the odds of comorbidity. A lower educational level and difficulty making ends meet, which largely fall in line with SES [64], showed up as significant risk factors for comorbidity. To some degree this result could again be explained by the mediating role of weight stigma. A large English cohort study by Hughes et al. (2024) showed that socioeconomically disadvantaged individuals experienced higher rates of internalised weight stigma irrespective of BMI [65]. Food insecurity, which disproportionately affects those with a low SES [66], could also play a part. Research shows that food insecurity is a driver for both the development of an ED [67, 68], and for having obesity/overweight [69]. Moreover, in low-income populations there are significant barriers to implementing consistent family meals, a protective factor in ED and obesity development [70, 71].

Lastly, this study provided evidence that having a mother from a non-EU country was also significantly associated with comorbidity. Ethnic minorities are more often categorised in the overweight or obesity category [72], and suffer from EDs at similar rates to white individuals [73]. However, they are particularly vulnerable to missed screening, diagnosis and treatment for EDs [53]. This oversight may stem from biases in clinical perception, rooted in the outdated “SWAG” stereotype, which assumes EDs predominantly affect “Skinny, White, Affluent Girls” [15, 74].

Himmelstein et al. (2017) examined the interplay between ethnicity, weight stigma, and DEB, revealing

that weight stigma impacts individuals across all ethnicities. However, responses to stigma vary: Hispanic women and black men were more likely, while black women were less likely, to exhibit DEB compared to white women [75]. These disparities may be partly explained by cultural differences in body image ideals. Belgium, like many EU countries, predominantly promotes the thin ideal [76], a standard shared by many Asian societies [77]. Conversely, African and Hispanic cultures often favour larger or curvier bodies [77]. Internalization of the thin ideal is a known risk factor for body dissatisfaction, which in turn increases the likelihood of EDs [34, 78]. Ethnic identity can mediate this effect; for instance, African ethnic identity has been associated with greater resistance to thin ideal internalization, whereas Asian identity appears to heighten susceptibility [73, 77, 79].

Although this study cannot draw definitive conclusions about ethnicity's role in the comorbidity of ED and obesity/overweight, the findings emphasize the importance of adopting an intersectional research approach (i.e. examining the interplay of embodied identities, including ethnicity, culture, and socioeconomic factors) [53].

Implications for the public health field

The public health field should leverage available data on the interrelatedness of higher weight and EDs [17–20]. For obesity prevention campaigns to be effective, it is crucial that they do not further stigmatise the individuals they are trying to help. Yet a systematic review by Bristow et al. (2020) shows that most public health campaigns are perceived to have the opposite effect [5]. As quoted by O'Dea (2004): “First, do no harm.” [80].

Regarding the evidence of this study, an interesting road to explore is the development of an integrative preventative approach tackling all disorders at once. For instance, Neumark et al. (2005) and Sanchez et al. (2011) suggest to regard obesity/overweight and EDs as being on the same side of a spectrum of weight-related disorders [81, 82]. They propose implementing overarching preventative programs that target shared risk and protective factors, such as discouraging dieting behaviour or promoting regular family meals [83].

An integrative approach could reduce costs and lower the chance of delivering contradictory or stigmatizing messages by incorporating insights from both ED and obesity research [81]. A recent systematic review by Le LK-D et al. (2022) identified several successful interventions, including the New Moves program—a school-based initiative combining principles from both fields [84]. The program focuses on behavioural changes rather than weight loss, with specialized physical education classes and activities aimed at improving self-image and encouraging healthy eating [85]. While some interventions showed promise in addressing both ED and

obesity outcomes, most, including New Moves, primarily improved ED-related outcomes such as body dissatisfaction and dieting behaviours, with limited effects on BMI [84]. The study concluded that further research is necessary to identify common risk factors and the most effective methods for jointly addressing EDs and high BMI [84]. A crucial step would be for more prevention programs to report variables related to both weight and EDs. Most studies report only one of both making them unfit to be included in a meta-analysis.

The Viasano programme, initiated in Belgium in 2007, is a community-based intervention grounded in the EPODE philosophy [86]. EPODE promotes healthy eating patterns and joyful daily movement, without stigmatizing any culture or behaviours. By mobilizing different local stakeholders surrounding families, it tries to tailor to the needs of all socioeconomic groups. Examples of previous actions of Viasano are organizing educational workshops on healthy food at schools, providing healthy lunches for children, and organizing sporting events in participating towns. Analyses of EPODE-based projects have shown a downward trend in BMI, but so far no outcomes related to ED prevalence have been recorded [87]. Expanding this project to include ED prevention measures, such as in the New Moves program [85], could potentially have a greater overall effect on ED and obesity/overweight prevalence. This could mean educating children and parents about the risk of dieting and weight talk, and promoting body satisfaction in children [56].

Besides an integrative approach, reframing obesity prevention campaigns to a general health-promoting policy could contribute to the health of the entire population, and lower the risk of stigmatising a large part of society that have overweight or obesity (currently half of the adult population in Belgium) [39, 88, 89]. For example, research by Puhl et al. (2013) shows that obesity-related health campaigns are more motivating when they promote specific health behaviours, such as eating healthy food and exercising, and do not mention the words ‘obese’ or ‘obesity’ [90]. Importantly, we also need to challenge the deep-rooted belief that having obesity is due to individual choice [28]. Public health messages should address unhealthy environments (such as marketing of fast foods, and availability and affordability of healthy foods) rather than lifestyle choices.

Furthermore, before implementing public health campaigns, it is necessary that they are reviewed for any stigmatizing language or visuals that possibly increase the risk of DEB or ED [80]. Although their stigmatizing potential remains controversial [82, 91], it is best to be cautious until consensus is reached. Preferred language regarding weight is subjective [92], so a user-centred approach involving the perspectives of ED experts, those with an ED, and those with higher weight when

developing public health campaigns is essential [88]. Additionally, we need to educate local stakeholders, such as health professionals, in non-stigmatizing language when addressing food, exercise and body image. Weight stigma is pervasive [28, 31, 57], and awareness of one's own implicit bias is therefore crucial. Health professionals should be trained to hold a high level of suspicion for EDs in people with higher weight, but also to enquire with care. Recently, the National Eating Disorders Collaboration (NEDC) in Australia published elaborate guidelines to assist health professionals in the assessment and management of ED in people of higher weight [92]. It supplies a list of warning signs, such as recent fluctuations in body weight, or requests for weight loss interventions, and recommends assessment tools with the strongest evidence for sensitivity, specificity and a low risk of stigmatisation, including the Eating Disorders Examination Questionnaire (EDE-Q). Roberts et al. (2023) provide some suggestions for the paediatric population, which could also be valuable for adults with higher weight. They advocate for a weight-neutral approach, implementing routine ED screening in all prepubescent children, and explaining to children with higher weight that such questions are asked of all patients [53].

Most importantly, any campaign promoting healthful behaviour will be in vain if the current infrastructure of unequal access to nutritious food and physical exercise is not addressed. Initial efforts of public health actors should be aimed at the environmental and commercial determinants of health. There is a need for structural policies that focus on the environment and not the individual [93, 94]. This means increasing affordability of healthy nutrition, creating opportunities and incentives for movement, and regulating food advertising. For example, one way to combat food insecurity across socioeconomic groups is providing free school meals [95], as recommended in the EU Child Guarantee [96]. A review paper by Guio et al. (2023) found that implementing school food programs not only decreased food insecurity, but also increased healthy eating behaviour and reduced the risk of having obesity/overweight [95]. However, as mentioned above, all public health measures must be carefully designed to avoid triggering DEB. An example that warrants caution is the rollout of mandatory calorie labelling on menus. This was intended to inform consumers about the healthiness of a certain menu item, but some experts have critiqued that it could potentially exacerbate ED symptoms [97]. Despite these remarks, increasing food literacy in the population remains an important public health goal. We should aim to supply information on the nutritional value of types of food in a non-moralising manner.

Strengths and limitations

This study is the first to examine the weight-related prevalence and odds of EDs in the Belgian population. It is based on a large and representative nationwide sample, therefore findings can be applied to inform future public health policies in Belgium.

While the SCOFF questionnaire is the most widely used ED screening test, it has been found to overestimate ED prevalence in general [98]. Although this is inherent to screening tests, it may have contributed to an overestimation of ED prevalence in our study. However, the diagnostic accuracy of the SCOFF is highest for young women with AN and BN, the specific conditions for which the test was originally developed and validated [45]. It is possible that the test may be less accurate when screening for newer DSM-V additions, such as BED or AAN, or individuals with higher weight categories [47]. This indicates that the actual prevalence and odds of EDs in higher-weight groups could be even greater than those reported in our study. In the future, it would be valuable to use a diagnostic tool that differentiates between various types of EDs more effectively.

This study was further limited by the use of self-reported height and weight data to calculate BMI. Research consistently shows self-reported height is typically overestimated, while weight is underestimated, leading to systematic underestimations of BMI [99]. Although objective measurements of weight and height are the gold standard, logistical and budgetary constraints often make this unfeasible for studies with large sample sizes. Nevertheless, applying correction factors to self-reported data could improve the accuracy of BMI classifications in future research.

Furthermore, the BHIS dataset lacked detailed information on ethnicity, which led us to use maternal country of birth as a proxy variable. This approach limited our ability to make definitive statements about the intersection of ED prevalence and ethnicity. More precise and detailed reporting on ethnicity in future studies would allow for a deeper investigation of this complex relationship.

Finally, the data for this study were collected in 2018, prior to the COVID-19 pandemic, which may have significantly influenced eating behaviours and related risk factors. Future research should examine whether the patterns observed in this study persist in more recent, post-pandemic data.

Conclusion

This study confirmed that, within the Belgian population, EDs disproportionately affect people with higher weight, compared to those with normal weight. EDs and higher weight co-occur more often in people with low

educational attainment, financial difficulties, and with maternal non-EU origin.

Although more longitudinal research is needed to find the exact causal links between a higher weight status and disordered eating behaviour, the present article provides robust evidence that public health campaigns take on an integrated preventative approach for all weight-related disorders.

Key action points for policymakers and healthcare practitioners:

1. Measure and report on variables related to both ED and weight outcomes, so integrative policies can be evaluated on effectiveness [82, 84].
2. Avoid messaging that solely targets weight loss: limit use of words such as 'obesity' or 'overweight', and focus on specific health behaviours [89, 90].
3. Focus critique on unhealthy environments rather than individual lifestyle choices [93, 94].
4. Assess existing and future health campaigns for their stigmatizing potential [28]. Involving people of higher weight, and collaboration between obesity and ED experts can help in this endeavour.
5. Educate health professionals to recognize ED symptoms in people of all weight categories, to ensure timely diagnosis of all patients. Train them to sensitively approach subjects like eating behaviour, physical exercise and body image, in a way that is free of stigmatization [88].
6. Address socioeconomic inequalities in access to healthy nutrition and physical exercise [71, 74].

Enacting these key points in public health policy will contribute to lower prevalence of EDs and DEB across all weight statuses.

This study enforces current literature, adds specific information about the Belgian population, and is one of the first studies to investigate variables associated with the comorbidity of ED and obesity or overweight.

Abbreviations

AN	Anorexia nervosa
AOR	Adjusted odds ratio
BED	Binge eating disorder
BHIS	Belgian health interview survey
BMI	Body mass index
BN	Bulimia nervosa
DEB	Disordered eating behavior
DSM-V	Diagnostic and statistical manual of mental disorders
ED	Eating disorder
EU	European union
OR	Odds ratio
SES	Socioeconomic status
SWAG	Skinny, white, affluent girls

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FG conceptualized this study together with RC. FG performed a literature review, wrote the original draft, and was aided in the analysis by RC. Revisions were made by RC, SD, LG and SV.

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Data availability

The datasets generated and/or analysed during the current study can be requested at <https://www.sciensano.be/en/use-health-interview-survey-2018-data>.

Declarations

Ethics approval and consent to participate

Informed consent was obtained from all subjects involved in the Health Interview Survey. The data collection was approved by the ethics committee of the Ghent university hospital (Commissie voor Medische Ethiek), Belgian Registration Number: B670201734213. All methods were carried out in accordance with declaration of Helsinki.

Consent for publication

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Competing interests

The authors declare no competing interests.

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References

1. WHO Consultation on Obesity (1999: Geneva, Switzerland) & World Health Organization. (2000). Obesity: preventing and managing the global epidemic: report of a WHO consultation. World Health Organization. Available from: <http://iris.who.int/handle/10665/42330>
2. Swinburn BA, Sacks G, Hall KD, McPherson K, Finegood DT, Moodie ML, et al. The global obesity pandemic: shaped by global drivers and local environments. *Lancet*. 2011;378(9793):804–14.
3. WHO. Obesity and overweight. 2021 [cited 2024 Feb 26]. Available from: <http://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight>
4. De Pauw R, Claessens M, Gorasso V, Drieskens S, Faes C, Devleesschauwer B. Past, present, and future trends of overweight and obesity in Belgium using bayesian age-period-cohort models. *BMC Public Health*. 2022;22(1):1309.
5. Bristow C, Meurer C, Simmonds J, Snell T. Anti-obesity public health messages and risk factors for disordered eating: a systematic review. *Health Promot Int*. 2020;35(6):1551–69.
6. Corsica JA, Hood MM. Eating disorders in an obesogenic environment. *J Am Diet Assoc*. 2011;111(7):996–1000.
7. Ramos Salas X. The ineffectiveness and unintended consequences of the public health war on obesity. *Can J Public Health*. 2015;106(2):e79–81.
8. American Psychiatric Association. Diagnostic and statistical manual of mental disorders. 5th ed. Washington, DC: American Psychiatric Association Publishing; 2022.
9. Guarda A. Eating disorders: Overview of epidemiology, clinical features, and diagnosis. In: UpToDate, Connor RF, editor, Wolters Kluwer. (Accessed on March 4, 2024).
10. Mehler P. Anorexia nervosa in adults and adolescents: Medical complications and their management. In: UpToDate, Connor RF, editor, Wolters Kluwer. (Accessed on March 4, 2024).
11. Chesney E, Goodwin GM, Fazel S. Risks of all-cause and suicide mortality in mental disorders: a meta-review. *World Psychiatry*. 2014;13(2):153–60.
12. Harris EC, Barraclough B. Excess mortality of mental disorder. *Br J Psychiatry*. 1998;173:11–53.

13. Nagata JM, Garber AK, Tabler JL, Murray SB, Bibbins-Domingo K. Prevalence and correlates of disordered eating behaviors among young adults with overweight or obesity. *J Gen Intern Med*. 2018;33(8):1337–43.
14. Bowman-Busato J, Schreurs L, Halford JCG, Yumuk V, O'Malley G, Woodward E, et al. Providing a common language for obesity: the European Association for the Study of Obesity obesity taxonomy. *Int J Obes (Lond)*. 2025;49(2):182–91. <https://doi.org/10.1038/s41366-024-01565-9>
15. Bruch H. Evolution of a psychotherapeutic approach to eating disorders: obesity, anorexia nervosa, and the person within. New York: Basic Books; 1973.
16. American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders. 3d ed. Washington, DC: American Psychiatric Association Publishing; 1980.
17. Al Banna MH, Brazendale K, Khan MSI, Sayeed A, Hasan MT, Kundu S. Association of overweight and obesity with the risk of disordered eating attitudes and behaviors among Bangladeshi university students. *Eat Behav*. 2021;40:101474.
18. Chen JY, Berner LA, Brown TA, Wierenga CE, Kaye WH. Associations of elevated weight status with symptom severity and treatment outcomes in binge/purge eating disorders. *Int J Eat Disord*. 2021;54(4):621–6.
19. da Luz FQ, Sainsbury A, Mannan H, Touyz S, Mitchison D, Hay P. Prevalence of obesity and comorbid eating disorder behaviors in South Australia from 1995 to 2015. *Int J Obes (Lond)*. 2017;41(7):1148–53.
20. Darby A, Hay P, Mond J, Quirk F, Buttner P, Kennedy L. The rising prevalence of comorbid obesity and eating disorder behaviors from 1995 to 2005. *Int J Eat Disord*. 2009;42(2):104–8.
21. Romano KA, Lipson SK, Beccia AL, Quatromoni PA, Gordon AR, Murgueitio J. Changes in the prevalence and sociodemographic correlates of eating disorder symptoms from 2013 to 2020 among a large National sample of U.S. Young adults: A repeated cross-sectional study. *Int J Eat Disord*. 2022;55(6):776–89.
22. Stunkard AJ. Eating patterns and obesity. *Psychiatr Q*. 1959;33:284–95.
23. Spitzer RL, Devlin MJ, Walsh BT, Hasin DS, Wing RR, Marcus MD, et al. Binge eating disorder: A multisite field trial of the diagnostic criteria. *Int J Eat Disord*. 1992;11:191–203.
24. Bulik CM, Sullivan PF, Kendler KS. Genetic and environmental contributions to obesity and binge eating. *Int J Eat Disord*. 2003;33(3):293–8.
25. Polivy J, Herman CP. Dieting and bingeing: A causal analysis. *Am Psychol*. 1985;40(2):193–201.
26. Fairburn CG, Doll HA, Welch SL, Hay PJ, Davies BA, O'Connor ME. Risk factors for binge eating disorder: A Community-Based, Case-Control study. *Arch Gen Psychiatry*. 1998;55(5):425–32.
27. Lebow J, Sim LA, Kransdorf LN. Prevalence of a history of overweight and obesity in adolescents with restrictive eating disorders. *J Adolesc Health*. 2015;56(1):19–24.
28. Rubino F, Puhl RM, Cummings DE, Eckel RH, Ryan DH, Mechanick JL, et al. Joint international consensus statement for ending stigma of obesity. *Nat Med*. 2020;26(4):485–97.
29. Cardel MI, Newsome FA, Pearl RL, Ross KM, Dillard JR, Miller DR, et al. Patient-Centered care for obesity: how health care providers can treat obesity while actively addressing weight stigma and eating disorder risk. *J Acad Nutr Diet*. 2022;122(6):1089–98.
30. Jackson SE, Beeken RJ, Wardle J. Perceived weight discrimination and changes in weight, waist circumference, and weight status. *Obes (Silver Spring)*. 2014;22(12):2485–8.
31. Puhl RM, Heuer CA. The stigma of obesity: a review and update. *Obes (Silver Spring)*. 2009;17(5):941–64.
32. Taylor SA, Ditch S, Hansen S. Identifying and preventing eating disorders in adolescent patients with obesity. *Pediatr Ann*. 2018;47(6):e232–7.
33. Golden NH, Schneider M, Wood C. Preventing obesity and eating disorders in adolescents. *Pediatrics*. 2016;138(3).
34. Neumark-Sztainer DR, Wall MM, Haines JI, Story MT, Sherwood NE, van den Berg PA. Shared risk and protective factors for overweight and disordered eating in adolescents. *Am J Prev Med*. 2007;33(5):359–69.
35. da Luz FQ, Hay P, Touyz S, Sainsbury A. Obesity with comorbid eating disorders: associated health risks and treatment approaches. *Nutrients*. 2018;10(7).
36. Balantekin KN, Grammer AC, Fitzsimmons-Craft EE, Eichen DE, Graham AK, Monterubio GE, et al. Overweight and obesity are associated with increased eating disorder correlates and general psychopathology in university women with eating disorders. *Eat Behav*. 2021;41:101482.
37. Villarejo C, Fernández-Aranda F, Jiménez-Murcia S, Peñas-Lledó E, Granero R, Penelo E, et al. Lifetime obesity in patients with eating disorders: increasing prevalence, clinical and personality correlates. *Eur Eat Disord Rev*. 2012;20(3):250–4.
38. Rodgers RF, Watts AW, Austin SB, Haines J, Neumark-Sztainer D. Disordered eating in ethnic minority adolescents with overweight. *Int J Eat Disord*. 2017;50(6):665–71.
39. Drieskens S, Charafeddine R, Gisle L, Gezondheidsenquête. 2018: Voedingsstatus. Brussel, België: Sciensano; 2019. Report No.: D/2019/14.440/53. Available from: www.gezondheidsenquête.be.
40. World Health Organization. WHO European Regional Obesity Report. 2022. Regional Office for Europe. License: CC BY-NC-SA 3.0 IGO. Available from: <http://iris.who.int/handle/10665/353747>
41. Goldblatt PB, Moore ME, Stunkard AJ. Social factors in obesity. *JAMA*. 1965;192(12):1039–44.
42. Huryk KM, Drury CR, Loeb KL. Diseases of affluence? A systematic review of the literature on socioeconomic diversity in eating disorders. *Eat Behav*. 2021;43:101548.
43. West CE, Goldschmidt AB, Mason SM, Neumark-Sztainer D. Differences in risk factors for binge eating by socioeconomic status in a community-based sample of adolescents: findings from project EAT. *Int J Eat Disord*. 2019;52(6):659–68.
44. Demarest S, Van der Heyden J, Charafeddine R, Drieskens S, Gisle L, Tafforeau J. Methodological basics and evolution of the Belgian health interview survey 1997–2008. *Archives Public Health*. 2013;71(1):24.
45. Morgan JF, Reid F, Lacey JH. The SCOFF questionnaire: assessment of a new screening tool for eating disorders. *BMJ*. 1999;319(7223):1467–8.
46. Garcia FD, Grigioni S, Chelali S, Meyrignac G, Thibaut F, Dechelotte P. Validation of the French version of SCOFF questionnaire for screening of eating disorders among adults. *World J Biol Psychiatry*. 2010;11(7):888–93.
47. House ET, Lister NB, Seidler AL, Li H, Ong WY, McMaster CM, et al. Identifying eating disorders in adolescents and adults with overweight or obesity: A systematic review of screening questionnaires. *Int J Eat Disord*. 2022;55(9):1171–93.
48. Sun Q, van Dam RM, Spiegelman D, Heymsfield SB, Willett WC, Hu FB. Comparison of dual-energy x-ray absorptiometric and anthropometric measures of adiposity in relation to adiposity-related biologic factors. *Am J Epidemiol*. 2010;172(12):1442–54.
49. Cole TJ, Bellizzi MC, Flegal KM, Dietz WH. Establishing a standard definition for child overweight and obesity worldwide: international survey. *BMJ*. 2000;320(7244):1240–3.
50. Eurostat. Key figures on European living conditions – 2024 edition. Luxembourg: Publications Office of the European Union; 2024. Available from: <http://ec.europa.eu/eurostat/documents/15216629/20161003/KS-01-24-001-E-N-N.pdf>
51. StataCorp. Stata statistical software: release 17. College station, TX: StataCorp LLC; 2021.
52. Duncan AE, Ziobrowski HN, Nicol G. The prevalence of past 12-Month and lifetime DSM-IV eating disorders by BMI category in US men and women. *Eur Eat Disord Rev*. 2017;25(3):165–71.
53. Roberts KJ, Chaves E. Beyond binge eating: the impact of implicit biases in healthcare on youth with disordered eating and obesity. *Nutrients*. 2023;15(8).
54. Sim LA, Lebow J, Billings M. Eating disorders in adolescents with a history of obesity. *Pediatrics*. 2013;132(4):e1026–30.
55. Chaves E, Jeffrey DT, Williams DR. Disordered eating and eating disorders in pediatric obesity: assessment and next steps. *Int J Environ Res Public Health*. 2023;20(17).
56. Hayes JF, Fitzsimmons-Craft EE, Karam AM, Jakubiak J, Brown ML, Wilfley DE. Disordered eating attitudes and behaviors in youth with overweight and obesity: implications for treatment. *Curr Obes Rep*. 2018;7(3):235–46.
57. Puhl R, Suh Y. Health consequences of weight stigma: implications for obesity prevention and treatment. *Curr Obes Rep*. 2015;4(2):182–90.
58. Puhl RM, Himmelstein MS, Pearl RL. Weight stigma as a psychosocial contributor to obesity. *Am Psychol*. 2020;75(2):274–89.
59. Sutin A, Robinson E, Daly M, Terracciano A. Weight discrimination and unhealthy eating-related behaviors. *Appetite*. 2016;102:83–9.
60. de Fortes Cavalcanti P, Brito E, de Magalhães Cunha C, Ribas de Farias Costa P, da Purificação Nazaré Araújo M, Cardoso Martins P et al. Weight stigma is a predictor of disordered eating in Brazilian college students during the COVID-19 pandemic: A 16-month cohort follow-up. *Appetite*. 2024;192:107084.

61. Pearlman AT, Murphy MA, Raiciulescu S, Gray JC, Klein DA, Schvey NA. The prospective relationship between weight-based discrimination and eating pathology among youth. *Eat Behav.* 2023;49:101746.
62. Sonnevile KR, Thurston IB, Gordon AR, Richmond TK, Weeks HM, Lipson SK. Weight stigma associated with mental health concerns among college students. *Am J Prev Med.* 2024;66(2):260–8.
63. Spahlholz J, Baer N, König HH, Riedel-Heller SG, Luck-Sikorski C. Obesity and discrimination - a systematic review and meta-analysis of observational studies. *Obes Rev.* 2016;17(1):43–55.
64. Australian Bureau of Statistics. (2011). Measures of socioeconomic status. (ABS Publication No. 1244.0.55.001). Available from: [http://www.ausstats.abs.gov.au/Ausstats/subscriber.nsf/0/367D3800605DB064CA2578B60013445C/\\$File/1244055001_2011.pdf](http://www.ausstats.abs.gov.au/Ausstats/subscriber.nsf/0/367D3800605DB064CA2578B60013445C/$File/1244055001_2011.pdf) [cited 2024 Mar 04].
65. Hughes AM, Flint SW, Clare K, Kousoulis AA, Rothwell ER, Bould H et al. Demographic, socioeconomic and life-course risk factors for internalized weight stigma in adulthood: evidence from an english birth cohort study. *Lancet Reg Health – Europe.* 2024;40.
66. Sciensano. Determinanten Van voedingskeuze: voedselzekerheid, voedselconsumptiepeiling 2022–2023. Juni; 2024.
67. Becker CB, Middlemass K, Taylor B, Johnson C, Gomez F. Food insecurity and eating disorder pathology. *Int J Eat Disord.* 2017;50(9):1031–40.
68. Hazzard VM, Loth KA, Hooper L, Becker CB. Food insecurity and eating disorders: a review of emerging evidence. *Curr Psychiatry Rep.* 2020;22(12):74.
69. Carvajal-Aldaz D, Cucalon G, Ordóñez C. Food insecurity as a risk factor for obesity: A review. *Front Nutr.* 2022;9:1012734.
70. Neumark-Sztainer D, Wall M, Fulkerson JA, Larson N. Changes in the frequency of family meals from 1999 to 2010 in the homes of adolescents: trends by sociodemographic characteristics. *J Adolesc Health.* 2013;52(2):201–6.
71. West CE, Sato AF. Weight and disordered eating among adolescents from low-income backgrounds. *Eat Behav.* 2023;50:101768.
72. Kumanyika SK. Unraveling common threads in obesity risk among racial/ethnic minority and migrant populations. *Public Health.* 2019;172:125–34.
73. Cheng ZH, Perko VL, Fuller-Marashi L, Gau JM, Stice E. Ethnic differences in eating disorder prevalence, risk factors, and predictive effects of risk factors among young women. *Eat Behav.* 2019;32:23–30.
74. Sonnevile KR, Lipson SK. Disparities in eating disorder diagnosis and treatment according to weight status, race/ethnicity, socioeconomic background, and sex among college students. *Int J Eat Disord.* 2018;51(6):518–26.
75. Himmelstein MS, Puhl RM, Quinn DM. Intersectionality: an understudied framework for addressing weight stigma. *Am J Prev Med.* 2017;53(4):421–31.
76. Bordo S. Unbearable weight: feminism, Western culture, and the body. Univ of California; 2023.
77. Rakhkovskaya LM, Warren CS. Ethnic identity, thin-ideal internalization, and eating pathology in ethnically diverse college women. *Body Image.* 2014;11(4):438–45.
78. Stice E. Risk and maintenance factors for eating pathology: a meta-analytic review. *Psychol Bull.* 2002;128(5):825.
79. Burke NL, Schaefer LM, Karvay YG, Bardone-Cone AM, Frederick DA, Schaumburg K, et al. Does the tripartite influence model of body image and eating pathology function similarly across racial/ethnic groups of white, black, latina, and Asian women? *Eat Behav.* 2021;42:101519.
80. O'Dea AJ. Prevention of child obesity: 'first, do no harm'. *Health Educ Res.* 2004;20(2):259–65.
81. Neumark-Sztainer D. Can we simultaneously work toward the prevention of obesity and eating disorders in children and adolescents? *Int J Eat Disord.* 2005;38(3):220–7.
82. Sánchez-Carracedo D, Neumark-Sztainer D, López-Guimerà G. Integrated prevention of obesity and eating disorders: barriers, developments and opportunities. *Public Health Nutr.* 2012;15(12):2295–309.
83. Neumark-Sztainer D. The interface between the eating disorders and obesity fields: moving toward a model of shared knowledge and collaboration. *Eat Weight Disord.* 2009;14(1):51–8.
84. Le LK-D, Tan EJ, Perez JK, Chiotelis O, Hay P, Ananthapavan J, et al. Prevention of high body mass index and eating disorders: a systematic review and meta-analysis. *Eating and weight Disorders - Studies on anorexia. Bulimia Obes.* 2022;27(8):2989–3003.
85. Neumark-Sztainer DR, Friend SE, Flattum CF, Hannan PJ, Story MT, Bauer KW, et al. New moves-preventing weight-related problems in adolescent girls a group-randomized study. *Am J Prev Med.* 2010;39(5):421–32.
86. Vinck J, Brohet C, Roillet M, Dramaix M, Borys JM, Beysens J, et al. Downward trends in the prevalence of childhood overweight in two pilot towns taking part in the VIASANO community-based programme in belgium: data from a National school health monitoring system. *Pediatr Obes.* 2016;11(1):61–7.
87. Borys JM, Le Bodo Y, Jebb SA, Seidell JC, Summerbell C, Richard D, et al. EPODE approach for childhood obesity prevention: methods, progress and international development. *Obes Rev.* 2012;13(4):299–315.
88. Treasure J, Ambwani S. Addressing weight stigma and anti-obesity rhetoric in policy changes to prevent eating disorders. *Lancet.* 2021;398(10294):7–8.
89. Hunger JM, Smith JP, Tomiyama AJ. An Evidence-Based rationale for adopting Weight-Inclusive health policy. *Social Issues Policy Rev.* 2020;14(1):73–107.
90. Puhl R, Peterson JL, Luedicke J. Fighting obesity or obese persons? Public perceptions of obesity-related health messages. *Int J Obes (Lond).* 2013;37(6):774–82.
91. Schwartz MB, Henderson KE. Does obesity prevention cause eating disorders?? *J Am Acad Child Adolesc Psychiatry.* 2009;48(8):784–6.
92. Ralph AF, Brennan L, Byrne S, Caldwell B, Farmer J, Hart LM, et al. Management of eating disorders for people with higher weight: clinical practice guideline. *J Eat Disord.* 2022;10(1):121.
93. Walls HL, Peeters A, Proietto J, McNeil JJ. Public health campaigns and obesity - a critique. *BMC Public Health.* 2011;11(1):136.
94. O'Hara L, Gregg J. The war on obesity: a social determinant of health. *Health Promot J Austr.* 2006;17(3):260–3.
95. Guio A-C. Free school meals for all poor children in europe: an important and affordable target? *Child Soc.* 2023;37(5):1627–45.
96. Guio A-C, Frazer H, Marlier E, editors. Study on the economic implementing framework of a possible EU child guarantee scheme including its financial foundation, second phase of the feasibility study for a child guarantee (FSCG2. Brussels: European Commission: Final Report; 2021.
97. Putra IGNE, Polden M, Wareing L, Robinson E. Acceptability and perceived harm of calorie labeling and other obesity policies: A cross-sectional survey study of UK adults with eating disorders and other mental health conditions. *Int J Eat Disord.* 2023;56(11):2049–61.
98. Kutz AM, Marsh AG, Gunderson CG, Maguen S, Masheb RM. Eating disorder screening: a systematic review and Meta-analysis of diagnostic test characteristics of the SCOFF. *J Gen Intern Med.* 2020;35(3):885–93.
99. Drieskens S, Demarest S, Bel S, De Ridder K, Tafforeau J. Correction of self-reported BMI based on objective measurements: a Belgian experience. *Archives Public Health.* 2018;76(1):10.

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